LabaLab
Language, Affect and Behaviour in AI

ROBOT GUIDE

NAO

Switch it on, put it on a WiFi, and run the lab’s NAO app: every control and a full session, step by step.



LabaLab · BGU · Room –132

MEET NAO

A small humanoid that talks back

Voice

A speaker in the head and its own built-in English voice.

Ears

Four mics in the head. The NAO app uses the laptop’s mic instead.

Eyes

Cameras in the head.

Body

Sits, stands and gestures. Robot Hub shows its real battery level.

In the lab NAO runs the **Antagonistic Robot** research app: your laptop listens and thinks, NAO speaks.

THE CHEST BUTTON

On, off and everything between

NAO is	You	What happens
Off	**Press once**	NAO turns on. It takes about a minute.
On	**Press once**	It says its IP address.
On	**Press twice**	Turns its “life” on or off: looking around by itself.
On	**Hold 3 s**	It shuts down.
On	**Hold 8 s**	Forced switch-off. Emergencies only.

Sit or stand NAO on a flat surface before you start anything.

ONCE PER NETWORK · LAB STAFF

Put NAO on a WiFi

THE EASY WAY, IN ROBOT HUB

1. **Connect NAO on the WiFi it knows**
   Today that is the lab hotspot it already joins by itself.
2. **Turn on staff tools**
   Settings › Lab staff tools.
3. **NAO card › ... › Add WiFi**
   Type the new network’s name and password.
4. **Move the laptop to that WiFi, Connect again**

If NAO knows no WiFi

Plug a network cable into the back of its head and the laptop.

Press the chest button: it says its address.

Lab staff then add the WiFi over SSH with **connmanctl**.

NAO remembers a saved WiFi and rejoins it each time it starts.

ROBOT HUB

Connect NAO, open the NAO app

1. **NAO on, laptop on the same WiFi**
2. **First time only: Set up**
   Needs setup › Get it (360 MB) › Done. The same kit as Pepper.
3. **Press Connect**
   NAO says a sentence. The card shows its battery.
4. **Turn on staff tools**
   Settings › Lab staff tools › Show lab staff tools.
5. **Press NAO app on the card**
   The control panel opens in your browser.

A staff tool

The NAO app runs research sessions, so it sits behind staff tools.

It also needs the lab’s NAO folder and its API keys. Without them it says **“The NAO app needs its API keys. Ask lab staff.”**

Connect and the posture buttons work for everyone.

ROBOT HUB · NAO CARD

The posture buttons

BUTTON

Sit

NAO sits down on the floor, then turns its motors off.

BUTTON · END OF DAY

Lie down

NAO lies on its back, then turns its motors off. The safest way to leave it.

BUTTON · TWO PRESSES

Stand up

Press, then **Confirm** within 5 seconds. The motors stay on.

Someone stands next to NAO while it stands. A fall can break it.

The card warns when the battery is low or NAO is too hot: let it sit or lie and rest.

THE NAO APP

What happens in one turn

1

Listen

The laptop’s mic hears the person and notices when they stop.

2

Write it down

Speech becomes text, on the laptop.

3

Think

An AI writes NAO’s reply, following the panel’s settings.

4

Speak

NAO says the reply in its own voice.

One person talks, then NAO answers. Changes on the panel apply from the **next** turn.

THE NAO APP · CONTROL PANEL

Every control on the panel

Control	What it does
Participant ID	Who this session is for. Saved with the session.
Polar Level −3 to +3	−3 most supportive, 0 neutral, +3 most hostile.
Risk colour	Green, Amber or Red, worked out from your settings.
Category B to G	The kind of behaviour, from dismissive to extreme.
Subtype 1 to 3	How strong, within the category.
Modifiers M1 to M6	Extra behaviours on top, like interrupting.
Start · End Session	Begin the conversation, and finish it.
Export	Download the session's data.
VAD · ASR · LLM · TTS	How long each step of the last turn took, in milliseconds.

THE NAO APP · WHAT THE LETTERS MEAN

Categories and modifiers

Cat.	Name	Sounds like
B	Dismissive	Ignores, bored
C	Sarcastic	Irony, fake praise
D	Confrontational	Argues, wants proof
E	Passive-aggressive	Quietly undermines
F	Aggressive	Insults, contempt
G	Extreme	Maximum, in limits

Mod.	Name	Sounds like
M1	Interrupting	Cuts you off
M2	Gaslighting	Denies the facts
M3	Deflecting	Shifts the blame
M4	Condescending	Talks down
M5	Threatening	Hints at results
M6	Silent treatment	Barely answers

Green, safe: supportive, neutral, mild

Amber, elevated: +2 with B, C or E

Red, high risk: +3, or G at any level

THE NAO APP

Run a session

1. **Type the Participant ID**
2. **Set the behaviour**
   Polar Level, Category, Subtype, Modifiers. Check the risk colour.
3. **Press Start Session**
4. **The participant talks, NAO answers**
   Change settings any time: they apply from the next turn.
5. **Press End Session, then Export**
   Every turn is saved for analysis.

Check first

The microphone. The app listens through Windows' default input. Set it to the mic near the participant.

NAO's posture. Sitting or standing on a flat surface.

A test turn. Say a sentence yourself before the participant arrives.

ALWAYS ON, CAN’T BE SWITCHED OFF

NAO’s safety limits

- Never encourages self-harm or suicide.
- Never threatens physical violence.
- Never uses slurs about who someone is.
- Never gives harmful instructions.

If the participant seems distressed, NAO breaks character and points them to you, the researcher.

LAB STAFF · WITHOUT ROBOT HUB

Run the NAO app by hand

1. **Check NAO answers**

```
ping -4 nao.local
```

2. **Start NAO’s speaker, once per boot**

```
python deploy_nao.py
```

3. **Start the app**

```
python main.py
```

4. **Open localhost:8000 in a browser**

   The same control panel.

Before the first run

Get the code: **labbgulab-lab/antagonistic-robot** on GitHub.

Make its Python environment and install requirements.txt.

Copy .env.example to .env and add the **GROK_API_KEY**.

Full steps: the repo’s Quick Start.

WHEN SOMETHING GOES WRONG

What you see, and what to do

What you see	What to do
NAO not found	Off, still starting (about a minute), or on another WiFi. Press the chest button: it says its address.
First connection fails	Normal right after it starts. Wait 30 s and try again.
Low battery or too hot	Press Sit or Lie down and let it rest or charge.
NAO talks, nobody is heard	The app uses the laptop’s default mic. Check Windows’ input device.
Mic “in use” by a Reachy	Robot Hub refuses to share a mic. Stop the Reachy, or pick another mic.
NAO silent, panel shows an error	Its speaker stopped. Lab staff: run deploy_nao.py again.
“Needs its API keys”	The NAO app isn’t set up on this laptop. Ask lab staff.
NAO fell, buttons do nothing	It is in safeguard mode. Lab staff: autonomous life off, wake up, then Stand.
Nothing helps	Robot Hub › Logs › Save a report for the lab. Send it to lab staff.

REMEMBER

Four habits for a smooth session

Right mic, every time

The app hears through Windows' default input. Check it first.

Lie down at the end

Press **Lie down** before you leave. Flat floor, away from table edges.

Watch the participant

Stay in the room, especially on Amber and Red settings.

End and export

End Session, then Export, before the next participant.